PROJECT: VILLA / BUILDING: **Villa Complex 415V LV Network**
DB No.: **DB-GF1** — Villa ground floor DB
FED FROM: MDB / SMDB: **SMDB-GF** / METER ENCLOSURE

LOAD DISTRIBUTION SCHEDULE

(3 - Phase, 415 V)

LOCATION OF DB: **Villa GF – store room**

RATING OF INCOMER	RATG. OF ELCB	SL. No.	CIR No.	MCB RTG. IN AMPS	CCT WIRE SIZE mm²	ECC WIRE SIZE mm²	ROOM / AREA	CONNECTED LOADS / POINTS															LOAD PER CIRCUIT - WATT			REMARKS
								LTG	EX.FAN	SH.S/O	13A S.S/O	13A T.S/O	W/H	FCU	SPUR O/L	ISOL	COOKER	S A/C	WAT.PUMP	OTH	C.FAN	15A S/O	R	Y	B	
WATT / UNIT								100	40	20	250		3000				6000	2500	750		80	1000				
100 A ISOLATOR TP+N	ELCB-1 25 A 100 mA 4P	1	**R1**	10	2.5	2.5	Living lighting	12															1200			
		2	**Y1**	6	2.5	2.5	Majlis lighting	10																1000		
		3	**B1**	6	2.5	2.5	External lighting	10																	1000	
		4	**Y2**	6	2.5	2.5	Kitchen lighting	8	1															840		
		5	**B2**	6	2.5	2.5	Dining lighting	6																	600	
	ELCB-2 40 A 30 mA 4P	6	**R3**	40	6	6	Cooker										1						6000			
		7	**Y3**	20	4	4	Guest bath water heater						1											3000		
		8	**B3**	20	4	4	Kitchen water heater						1												3000	
		9	**R4**	16	4	4	Guest bedroom split A/C											1					2500			
		10	**Y4**	16	4	4	Living split A/C											1						2500		
		11	**B4**	16	4	4	Majlis split A/C											1							2500	
	ELCB-3 25 A 30 mA 4P	12	**R5**	10	4	4	Guest bedroom	4			4										1		1480			
		13	**Y5**	16	4	4	Living sockets				8													2000		
		14	**B5**	16	4	4	Dining split A/C											1							2500	
		15	**R6**	6	4	4	Maid room	3	1		3												1090			
		16	**Y6**	16	4	4	Laundry															2		2000		
		17	**B6**	10	4	4	Majlis sockets				6														1500	
	ELCB-4 25 A 30 mA 4P	18	**R7**	6	4	4	Water pump												1				750			
		19	**Y7**	6	4	4	Dining sockets				4													1000		
		20	**B7**	10	4	4	Kitchen sockets				5														1250	
		21	**Y8**	6	4	4	Guest bathroom	2	1	1														260		
TOTAL CONNECTED LOAD (W) — the circuits above																							**13020**	**12600**	**12350**	**37970 W**
MAXIMUM DEMAND (W) — after demand factors																							**13020**	**12600**	**12350**	**37970 W · imbalance 5 %**

CABLE SIZE: 4C 35mm² Cu XLPE/SWA/PVC + 1C 16mm² Cu/PVC G/Y AS ECC, 30 m

C.FAN = Ceiling Fan, EX.FAN = Exhaust Fan, SH. S/O = Shaver Socket-outlet, W/H = Water Heater, H/D = Hand Dryer, 'W' = Window type & 'S' = Split type. Circuit references: R/Y/B = phase, number = way; RYB = 3-phase circuit.